Tutorial Worksheet 3

AI 501: Mathematics for Artificial Intelligence | Fall 2026

Problem 1

A tutor records how many hours three students studied (x) and their quiz marks (y):

Hours studied (x)	0	1	2
Mark (y)	2	3	7

We want to model this with a straight line, $\hat{y} = w_0 + w_1 x$.

(a) Write the model for all three students as a single matrix equation $Xw = y$. State X, w, and y, and their shapes.

(b) Is there a w that satisfies $Xw = y$ exactly? Why or why not?

(c) Find the w that minimizes the sum of squared errors, $\sum_i (y_i - \hat{y}_i)^2 = \|y - Xw\|^2$. Show your working.

(d) Predict the mark for a student who studies 3 hours.

Problem 2

A sensor is calibrated by recording its output y at three input voltages:

Voltage (x)	-1	0	1
Output (y)	3	1	1

An engineer decides to be generous with flexibility and fits a cubic:

$$\hat{y} = w_0 + w_1 x + w_2 x^2 + w_3 x^3.$$

(a) Write down the design matrix X and the vector w. What are their shapes?

(b) The engineer tries $w = (X^T X)^{-1} X^T y$. What is the shape of $X^T X$? Explain why this inverse does not exist. (*Hint:* what is the largest rank X can have?)

(c) Verify that $w = (1, -1, 1, 0)^T$ fits the data exactly. Then show that $w = (1, 0, 1, -1)^T$ also fits exactly.

(d) What is the highest degree M for which $(X^T X)^{-1}$ is guaranteed to exist for these three points? Find the resulting fit.